RescueNav: AI-Powered AAV Trajectory Planning With LoRa Connectivity for IoT-Based Disaster Communications

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***Abstract*—In disaster-affected areas where conventional communication infrastructure collapses, autonomous aerial vehicles (AAVs) can act as mobile IoT base stations to restore connectivity. However, maintaining reliable low-power wide-area links such as long range (LoRa) remains challenging due to terrain-dependent propagation, interference, and energy constraints. This work introduces RescueNav, an intent-driven AAV trajectory planning framework that integrates LoRa-specific signal constraints with reinforcement learning-based decision-making. Unlike existing methods that treat communication as a secondary factor, RescueNav formalizes the operator's intent to maintain line-of-sight connectivity, maximize coverage, and conserve AAV energy directly into the trajectory optimization process. By leveraging Q-learning with prioritized experience replay (PER), RescueNav dynamically adapts flight paths to evolving disaster environments, guided by a shaped reward function that encodes communication and energy objectives. Experiments with a Holybro S500 V2 platform equipped with a Pixhawk, Raspberry Pi, and LoRa modules demonstrate that RescueNav improves median signal strength by 4×, enhances throughput by 42%, and achieves a 55% higher cumulative reward than standard Q-learning. These results highlight the potential of intent-based reinforcement learning (RL) for resilient IoT communication in mission-critical post-disaster scenarios.**

***Index Terms*—Autonomous aerial vehicle (AAV), intent-based networking, long range (LoRa), post-disaster relief, reinforcement learning (RL), trajectory optimization.**

I. Introduction

MASSIVE natural disasters such as earthquakes, floods, tsunamis, wildfires, and hurricanes result in widespread devastation, including loss of life, economic setbacks, and infrastructure collapse. Between 1970 and 2023, global economic losses from such events increased by more than 400% due to service disruptions and recovery challenges [1]. Reliable communication plays a crucial role in effective disaster response, yet damaged infrastructure often leads to complete isolation of affected regions [2], [3].

Post-disaster communication systems must be energy-efficient, easily deployable, and resilient to harsh environmental variations [14]. Autonomous aerial vehicles (AAVs) have emerged as an effective alternative due to their agility, rapid deployment, and low operational costs. Acting as temporary aerial base stations, AAVs can provide coverage over damaged regions while aligning with intent-based networking principles [12], [17], [21], [22]. In this context, intent-based networking enables AAVs and IoT devices to translate high-level mission requirements, such as "maintain connectivity" or "minimize delay," into optimized, actionable configurations for disaster recovery missions.

Low-power wide-area network technologies, especially long range (LoRa), offer LoRa communication with low energy consumption, making them suitable for infrastructure-limited post-disaster contexts [9], [10], [11]. However, AAV-based LoRa networks face terrain-driven signal variability, interference, and limited onboard power [25], [27]. While prior LoRa-based AAV studies, including [21] and [27], focus on antenna behavior or meta reinforcement learning (RL), they treat communication reliability and trajectory optimization as separate tasks. This separation leads to reactive behavior that cannot maintain stable connectivity in dynamic environments.

This work introduces RescueNav, a unified LoRa-driven AAV trajectory planning framework that embeds line-of-sight conditions and terrain information directly into the RL process. Unlike prior methods, RescueNav ensures that LoRa connectivity guides the trajectory generation process rather than serving as an aftereffect.

At the core of RescueNav is a lightweight RL-based LoRa trajectory planner that uses Q-learning with prioritized experience replay (PER), enabling real-time learning of energy and connectivity-aware policies within the computational limits of onboard hardware. A dynamic action space provider (DASP) further adapts decision-making by adjusting available actions based on obstacle density and local signal variation.

The key contributions of this work are summarized as follows.

1) A LoRa-driven trajectory planning framework that incorporates LoRa propagation and terrain constraints to

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ensure resilient connectivity within intent-based networking principles.
2) A lightweight Q-learning model enhanced with PER for real-time adaptation within onboard computational limits.
3) A DASP that adapts the AAV action space based on local obstacle density (LOD) and signal variations.
4) Real-world validation through extensive simulations and field experiments using a Holybro S500 V2 equipped with Pixhawk 6C, Raspberry Pi 4B, and LoRa modules.

By unifying communication reliability, energy awareness, and intent-driven optimization into a single RL framework, RescueNav advances beyond prior LoRa AAV studies and provides a scalable, field-tested solution for resilient AI-powered disaster communication systems.

II. Related Works

Research on AAV-assisted communication systems for disaster response has advanced across several areas, including energy efficiency, network resilience, intelligent coordination, and real-time adaptability. In [2], the importance of dynamic sensing and adaptable communication is emphasized, laying the foundation for trajectory optimization explored in [3] and [14]. These studies demonstrate how AAVs can manage diverse communication demands through adaptive resource allocation and rapid reconfiguration under degraded network conditions, aligning with the principles of intent-based networking.

Trajectory planning remains a central focus. Pan et al. [3] proposed energy-aware path algorithms to ensure continuous communication during emergencies, while Zhang et al. [4] presented a comprehensive optimization framework that integrates altitude control and link-quality constraints. Hierarchical clustering in [14] and federated RL in [9] improve base station efficiency and energy-aware coordination, respectively. Similarly, Saif et al. [8] introduced energy harvesting and clustering to extend mission duration in resource-scarce environments.

Decentralized control and system stability are explored in [6], which combines energy harvesting with cluster head selection to enhance network longevity. Chen et al. [5] introduced multiagent deep RL (MADRL) for distributed trajectory and power control, which was later extended in [11] to optimize power and path jointly. Robustness in multihop AAV networks under disaster-induced failures is further addressed in [7].

Recent works employ adaptive learning to enhance AAV decision-making in AI-powered IoT systems. In energy-constrained IoT contexts, Jabbari et al. [9] apply federated RL, while Du et al. [26] employ AI-based trajectory optimization under environmental uncertainty. Obstacle-rich environments are handled in [25] by compressing spatial information to accelerate RL convergence. Drone-hosted computation for emergency response is explored in [20], illustrating AAVs' dual role as communication relays and mobile edge nodes.

LoRa-aware AAV trajectory planning, however, remains underexplored. Zhang et al. [21] introduced a directivity-aware antenna model to enhance LoRa link quality through adaptive orientation, and Ghazali et al. [10] provided a survey of LoRa-based AAV deployments across urban and rural disaster zones. Yet, these works primarily focus on physical-layer optimization or static deployments, without incorporating LoRa propagation effects into trajectory learning or real-time control. Likewise, Soorki et al. [27] applied meta-RL to LoRa AAV adaptability but did not couple communication dynamics with path optimization, leading to reactive rather than predictive behavior.

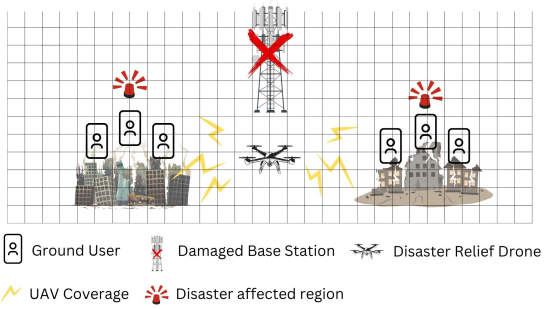


Fig. 1. Post-flood disaster scenario for AAV-assisted LoRa emergency communication. Terrain includes partially submerged buildings, user clusters in open spaces, and rugged aerial paths.

Additional studies explore situational adaptation in disaster response. AI-enabled autonomous drones for crisis assessment [17] accelerate environmental data collection. In contrast, GAN-based victim detection [22] and multi-AAV coordination frameworks [23], [24], [28] aim to balance latency, energy, and mobility tradeoffs. RL methods such as deep Q-networks (DQNs) and deep deterministic policy gradient (DDPG) [5] demonstrate strong potential for high-dimensional AAV control but often require substantial computational resources, limiting their feasibility on embedded AAV platforms.

In contrast, RescueNav unifies the capabilities of existing approaches into a lightweight and integrated framework. By explicitly modeling LoRa propagation, incorporating line-of-sight (LoS) awareness and terrain sensitivity within the learning loop, and applying Q-learning with PER, it enables real-time, energy-aware decision-making on low-power AAV hardware. The framework thus advances LoRa-based AAV communication from reactive adaptation toward proactive, intent-driven trajectory optimization suited for uncertain post-disaster environments.

III. System Model and Problem Formulation

A. System Model

As illustrated in Fig. 1, we consider a post-flood disaster environment where conventional ground-based stations are nonfunctional due to submersion or infrastructural collapse. Such events, common in coastal or river-adjacent urban zones, create fragmented user distribution, collapsed structures, and unpredictable terrain. To restore communication, a AAV equipped with LoRa-based wireless capability is deployed to serve isolated user equipment (UE) on the ground, as illustrated by the RescueNav system architecture in Fig. 2.

The AAV operates on a discretized 2-D grid overlaid on the disaster region, with each cell representing a fixed-size area. The environment is partially known from pre-disaster

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TABLE I
ENVIRONMENTAL VARIABLES AND CONSTRAINTS

Parameter	Value / Unit
Grid Size ($G \times G$)	20×20 cells, $25\,\text{m} \times 25\,\text{m}$
Obstacle Density (O_d)	0–100% (non-flyable cells)
User Mobility (U_m)	Probability $[0, 1]$ per step
Link Quality (S_t)	RSSI (dBm) / SNR (dB)
Energy Budget ($E_{\max}$)	1000 mAh (battery capacity)
Move Energy (E_m)	25 mAh per move
Altitude Bins	$\{10, 20, 30\}$ m
Zone Priority (Z_p)	Integer $\{1, 2, 3\}$ (low to high)
Dynamic Obstacles	Binary mask: 1 = blocked, 0 = clear

maps and elevation models, though obstacles may shift due to rising water or debris. The AAV must traverse this space, avoid no-fly zones such as high-rise clusters or submerged regions, and select routes that maintain LoS links to ground users. Movement between adjacent cells incurs a fixed energy cost E_m, and the AAV is constrained by a limited battery capacity $E_{\max}$, including a reserve for safe return to the charging station.

The LoRa communication model follows a distance-based path-loss profile with elevation-dependent attenuation. Environmental factors such as obstacle density, user mobility, signal blockage from structures or water, and altitude-based LoRa link quality influence reliability. User distribution is spatially nonuniform, with clusters near relief centers, open areas, or structural remnants. These constraints and environmental parameters are summarized in Table I.

B. Problem Formulation

The objective is to determine a real-time AAV trajectory that maximizes ground UE coverage while maintaining LoRa LoS communication and satisfying all environmental and energy-related constraints in a post-flood setting. The AAV receives as input the segmented aerial map, which includes flyable and obstructed zones, a user priority heatmap, discrete altitude bins, and terrain-aware LoRa signal maps.

At each time step, the AAV observes its current state $s_t = (\text{location}, \text{altitude_bin}, \text{zone_index})$ and selects an action a_t corresponding to movement or hovering. The trajectory planning is modeled as an RL problem, where a policy π is trained to maximize the expected long-term reward over time

$$\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t R_t \,\middle|\, \pi\right] \tag{1}$$

subject to

$$\sum E_m \le E_{\max}, \quad S_t \ge S_{\text{thresh}}, \quad \frac{1}{T}\sum_{t=1}^{T} C_t \ge C_{\text{target}}, \quad O_t = 0$$

where

1) R_t reward at time t based on coverage gain, LoS quality, and zone priority;
2) γ discount factor for future rewards;
3) C_t number of UEs covered at time t;
4) S_t LoRa signal strength at time t';
5) O_t obstacle collision flag (must remain zero).

This formulation enables the AAV to adaptively navigate dynamic post-disaster terrains, maximizing coverage while maintaining communication quality and conserving energy for safe operation.

IV. PROPOSED MODEL

A. Preliminaries

The post-disaster scenario involves a AAV navigating an uncertain environment with variable user distribution and complex terrain. Missions begin at the charging station and end upon return, with the focus on optimizing the trajectory for maximum communication coverage. The MDP uses a finite-horizon setup with $T = 500$ steps per episode. Termination occurs upon energy depletion, returning to the station, or reaching the step limit. The states, actions, and rewards are defined as follows.

1) States: At any given time t, the state of the system is represented as $s_t = (x_t, y_t, z_t, I_{E_{\text{remain}}}, S_l)$, which includes the AAV's current position in 3-D space, specified by the grid coordinates (x_t, y_t) and altitude z_t, a binary indicator $I_{E_{\text{remain}}}$, which denotes whether the AAV's current energy level E_{remain} is above the threshold required to safely reach the charging station and the LoRa signal strength at time t, denoted as S_l.

2) Actions: At each time step t, the AAV selects an action from the discrete action space $A_t = \{A_1, A_2, \ldots, A_8\}$, where the actions correspond to: serving users in the current grid (A_1), moving east ($x+1, y, z$; A_2), south ($x, y-1, z$; A_3), west ($x-1, y, z$; A_4), north ($x, y+1, z$; A_5), ascending vertically ($x, y, z+1$; A_6), descending ($x, y, z-1$; A_7), or returning to the charging station (A_8).

3) Rewards: To guide the drone in maintaining a strong LoS for LoRa signal optimization and to maximize coverage, a reward function R is defined that combines LoS maintenance, obstacle avoidance, signal strength, energy efficiency, and task completion. The reward function is expressed as follows:

$$R = R_{\text{LoS}} + R_{\text{Obstacle}} + R_{\text{Signal}} + R_{\text{Energy}}.$$

Here, R_{LoS} provides a positive reward $+r_{\text{LoS}}$ if the LoS to the base station is maintained, and zero otherwise. R_{Obstacle} imposes a penalty $-r_{\text{Obstacle}}$ if an obstacle is encountered, and zero otherwise. R_{Signal} offers a reward $+r_{\text{Signal}}$ when signal strength is optimized, a penalty $-r_{\text{Signal}}$ if signal strength is degraded, and zero otherwise. Finally, R_{Energy} assigns a reward $+r_{\text{Energy}}$ when energy-efficient behavior is observed, and zero otherwise.

4) Learning Strategy: Q-learning with PER is selected for its sample efficiency, low memory requirements, and stability under nonstationary conditions. Compared to DQN or DDPG, Q-learning with PER achieves faster convergence to higher cumulative rewards in simulated ablation studies under LoRa constraints.

B. ALoS Optimization for LoRa Communication

The RescueNav framework begins with the adaptive LoS (ALoS) Optimization module, which utilizes tabular Q-learning with PER to ensure reliable LoRa communication, as formalized in Algorithm 1.

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1) State and Action Representation: At each time step t, the AAV is in a state $s_t = (x_t, y_t, z_t, I_{E_{\text{remain}}}, S_l)$, denoting position, remaining energy, and signal strength. The discrete action space is

$$A_t = \{A_1, A_2, \ldots, A_8\}$$

where A_1: serve users, A_2–A_5: move in four horizontal directions, A_6: ascend, A_7: descend, and A_8: return to base.

2) Learning Framework: A Q-table $Q(s, a)$ stores expected returns for action a in state s. At each step, action a_t is selected via ϵ-greedy exploration. Upon executing a_t, the AAV observes reward r_t, next state s', and signal strength change.

3) Reward Shaping: To incorporate signal quality, the total reward is defined as follows:

$$r_{\text{shaping}} = \alpha \left(\text{signal}_{\text{new}} - \text{signal}_{\text{old}}\right) \tag{2}$$

$$r_{\text{total}} = r_t + r_{\text{shaping}}. \tag{3}$$

Here, α scales the influence of signal variation on learning.

4) Q Value Update: The TD target is computed as follows:

$$y_i = \begin{cases} r_i, & \text{if terminal} \\ r_i + \gamma \max\limits_{a'} Q\left(s_i', a'\right), & \text{otherwise} \end{cases} \tag{4}$$

and the Q value is updated via

$$Q(s_i, a_i) \leftarrow Q(s_i, a_i) + \alpha \left(y_i - Q(s_i, a_i)\right). \tag{5}$$

5) Prioritized Experience Replay: To accelerate convergence, transitions are stored with priority

$$p_i = \alpha \cdot \text{relevance}(s_i, a_i). \tag{6}$$

Transitions with higher TD errors are more likely to be replayed, enhancing learning from impactful events.

6) Exploration Decay: The exploration rate decays over time

$$\epsilon \leftarrow \max\left(\epsilon_{\min}, \epsilon \cdot \epsilon_{\text{decay}}\right). \tag{7}$$

7) Convergence: The algorithm iterates until the Q-table stabilizes. The AAV gradually learns policies that ensure high signal strength, safe energy levels, and optimal LoS, without requiring function approximation.

C. Efficient Area Coverage With Exploration

Once LoRa connectivity is established, RescueNav activates a grid-based exploration policy using tabular Q-learning with experience replay. The AAV aims to maximize coverage across disconnected regions while avoiding obstacles and minimizing redundancy. The environment is modeled as a 2-D grid where each cell (i, j) is either free or obstructed. The AAV's state $s_t \in \mathbb{Z}^2$ denotes its grid position, and actions $a_t \in \{\text{up}, \text{down}, \text{left}, \text{right}\}$ move to adjacent unblocked cells. The reward is computed as $r_t = \beta \cdot \text{coverage}(s_t) - \lambda \cdot \text{penalty}(s_t, a_t)$, where $\text{coverage}(s_t) = 1$ if the cell is newly visited, and penalty = 1 for collisions or idling. We use $\beta = 10$ and $\lambda = 5$.

To ensure obstacle avoidance and signal-aware planning, a depth map $D(i, j)$ and signal strength map $S(i, j)$ guide movement. Signal strength is modeled as follows:

$$S(i, j) = \frac{\text{signal_intensity}}{\sqrt{(x_{\text{AAV}} - i)^2 + (y_{\text{AAV}} - j)^2 + \Delta h^2}}$$

where $\Delta h = z_{\text{AAV}} - D(i, j)$. Cells with $S(i, j) < 30\,\text{dBm}$ are considered interference-prone.

An ϵ-greedy policy manages exploration, with $\epsilon_{t+1} = \max(\epsilon_{\min}, \epsilon_t \cdot \epsilon_{\text{decay}})$, using $\epsilon_0 = 1.0$, $\epsilon_{\text{decay}} = 0.99$, and $\epsilon_{\min} = 0.05$.

Transitions (s_t, a_t, r_t, s_{t+1}) are stored in a circular replay buffer R of size 10 000. The Q values are updated using minibatches of size 32 with

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_{a'} Q\left(s_{t+1}, a'\right) - Q(s_t, a_t)\right]$$

where $\alpha = 0.1$ and $\gamma = 0.95$.

Training proceeds for 500 episodes or until 95% of reachable grid cells are explored. Simulations were run on a 100×100 grid (15% obstacle density) in MATLAB and Python. LoRa signal decay followed inverse-distance modeling. Field tests used an SL53VLX1 LiDAR for depth sensing and a Raspberry Pi 4B for onboard Q-table updates. Stability was ensured via accurate indexing, obstacle masking, and loop prevention. The complete procedure is summarized in Algorithm 2.

D. Dynamic Trajectory Optimization for Enhanced AAV Communication

RescueNav integrates the DASP with the ALoS module to generate real-time, communication-aware AAV trajectories, as mentioned in Algorithm 3.

The system adjusts flight direction and altitude based on obstacle density and signal attenuation, ensuring stable coverage in cluttered disaster zones. At each planning step, DASP evaluates LOD via LiDAR within a 5 m radius. A normalized quarter-division factor $qd \in [0, 4]$ determines available directions

$$D = 4 + 4 \cdot qd.$$

Sparse regions restrict motion to cardinal directions, while dense areas allow full 360° maneuvering. The local movement space (LMS) cone constrains feasible directions.

with angular width

$$\theta_{\text{LMS}} = 45^\circ + \left(\frac{\text{LOD}}{\text{LOD}_{\max}} \times 315^\circ\right).$$

This supports conservative motion in open areas and greater flexibility near obstacles. Candidate directions within the LMS cone are ranked by free-space availability and exploration progress. The ALoS module complements DASP by optimizing altitude from LoRa signal strength

$$S(\mathbf{s}_i, h) = \frac{P_t}{L(d_i, h) + L_{\text{obstacles}}}$$

where d_i is the distance to the nearest node. Signals below 30 dBm trigger rerouting or altitude changes. If obstacles $\mathcal{O}(\mathbf{s})$ block a path segment π_t, an alternative π_t' is computed to maintain connectivity. The AAV trajectory $\mathcal{T}$ is incrementally built using signal-compliant, obstacle-free segments prioritizing unexplored zones, followed by a return-to-base route. Field tests with an SL53VLX1 LiDAR and Raspberry Pi 4B showed stable real-time navigation under varying obstacle and signal conditions.

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Algorithm 1 ALoS Optimization of LoRa Signals via Q-Learning With PER

Input: State space S, action space A, learning rate α, discount factor γ, exploration parameters $(\epsilon_{\text{initial}}, \epsilon_{\min}, \epsilon_{\text{decay}})$, priority memory size N, shaping weight λ
Output: Optimized Q-table $Q(s,a)$ for UAV LoRa-aware navigation

Initialize $Q(s,a) \leftarrow 0, \forall s \in S, a \in A$;
Initialize replay memory D with capacity N;
Set episode counter $e \leftarrow 0$, exploration rate $\epsilon \leftarrow \epsilon_{\text{initial}}$;
while *not converged* **do**
  Reset UAV environment; observe initial state s_0;
  Set $t \leftarrow 0$, $done \leftarrow$ False;
  while $done$ *is False and* $t < T$ **do**
    Select action a_t using ϵ-greedy policy:

$$a_t = \begin{cases} \text{random action}, & \text{w.p. } \epsilon \\ \arg\max_a Q(s_t, a), & \text{otherwise} \end{cases}$$

    Execute a_t, observe next state s_{t+1}, base reward r_t, LoRa signal strength ϕ_{t+1}, and termination flag;
    Compute reward shaping:

$$r_{\text{shaped}} = r_t + \lambda \cdot (\phi_{t+1} - \phi_t)$$

    Compute temporal difference (TD) error:

$$\delta_t = r_{\text{shaped}} + \gamma \cdot \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)$$

    Store transition $(s_t, a_t, r_{\text{shaped}}, s_{t+1}, done)$ in D with priority $|\delta_t|$;
    Sample minibatch $\{(s_i, a_i, r_i, s_i', done_i)\}$ from D using prioritized sampling;
    foreach *sample* $(s_i, a_i, r_i, s_i', done_i)$ **do**
      if *done* **then**
        $y_i \leftarrow r_i$
      else
        $y_i \leftarrow r_i + \gamma \cdot \max_{a'} Q(s_i', a')$
      end
      Update:

$$Q(s_i, a_i) \leftarrow Q(s_i, a_i) + \alpha \cdot (y_i - Q(s_i, a_i))$$

    end
    Update $\epsilon \leftarrow \max(\epsilon_{\min}, \epsilon \cdot \epsilon_{\text{decay}})$;
    Set $s_t \leftarrow s_{t+1}$; increment t;
  end
  Increment $e \leftarrow e + 1$;
end

Algorithm 2 Efficient Coverage Maximization With Adaptive Exploration (ECMAE)

Input: State space S, action space A, actor parameters θ^μ, critic parameters θ^Q, discount factor γ, soft update rate τ, replay buffer $\mathcal{D}$, exploration noise $\mathcal{N}$
Output: Optimal UAV position maximizing coverage $s^* \in S$

Initialize actor network $\mu(s|\theta^\mu)$ and critic $Q(s,a|\theta^Q)$;
Initialize target networks μ' and Q' with $\theta^{\mu'} \leftarrow \theta^\mu$, $\theta^{Q'} \leftarrow \theta^Q$;
Initialize replay buffer $\mathcal{D}$, max coverage $C_{\max} \leftarrow 0$, optimal position $s^* \leftarrow s_0$;
for *each episode* **do**
  Initialize UAV to starting state s_0; set $done \leftarrow$ False;
  while *not done* **do**
    Select action $a_t = \mu(s_t|\theta^\mu) + \mathcal{N}_t$;
    Execute a_t, observe s_{t+1}, reward r_t, and coverage area $C(s_{t+1})$;
    Store transition (s_t, a_t, r_t, s_{t+1}) in $\mathcal{D}$;
    Sample minibatch (s_i, a_i, r_i, s_i') from $\mathcal{D}$;
    Compute target:

$$y_i = r_i + \gamma Q'\left(s_i', \mu'\left(s_i'|\theta^{\mu'}\right)|\theta^{Q'}\right)$$

    Update critic:

$$\theta^Q \leftarrow \theta^Q - \alpha_Q \nabla_{\theta^Q} \left(y_i - Q\left(s_i, a_i|\theta^Q\right)\right)^2$$

    Update actor using sampled policy gradient:

$$\nabla_{\theta^\mu} J \approx \mathbb{E}_{s_i}\left[\nabla_a Q\left(s_i, a|\theta^Q\right)|_{a=\mu(s_i)} \cdot \nabla_{\theta^\mu} \mu(s_i|\theta^\mu)\right]$$

    Soft-update target networks:

$$\theta^{Q'} \leftarrow \tau\theta^Q + (1-\tau)\theta^{Q'}, \quad \theta^{\mu'} \leftarrow \tau\theta^\mu + (1-\tau)\theta^{\mu'}$$

    if $C(s_{t+1}) > C_{\max}$ **then**
      $C_{\max} \leftarrow C(s_{t+1})$, $s^* \leftarrow s_{t+1}$
    end
    $s_t \leftarrow s_{t+1}$
  end
end
return s^*

E. System Integrity and Implementation

RescueNav ensures secure, reproducible deployment in disaster environments through a clear threat and mitigation framework. The system incorporates an explicit mapping between potential attack vectors and corresponding defenses to maintain communication integrity throughout AAV operation. Eavesdropping risks are mitigated through LoRa spread-spectrum modulation, which reduces the feasibility of interception across LoRa links. Spoofing attempts are addressed using cryptographic authentication between the AAV and the base station, ensuring that only verified control messages influence flight decisions. Signal integrity disruptions are managed through per-waypoint link verification, where the AAV evaluates packet quality and signal strength before executing trajectory updates. These mechanisms collectively establish a layered protection model while preserving interoperability with the LoRaWAN communication stack for telemetry exchange and integration within disaster-response infrastructures. The implementation uses Python 3.8, OpenAI Gym for environment setup, NumPy for Q-learning, and

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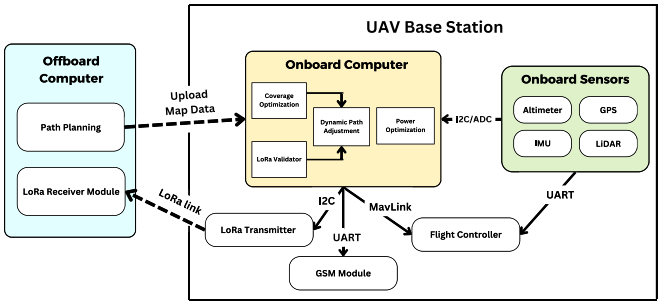


Fig. 2. Architecture diagram of RescueNav framework.

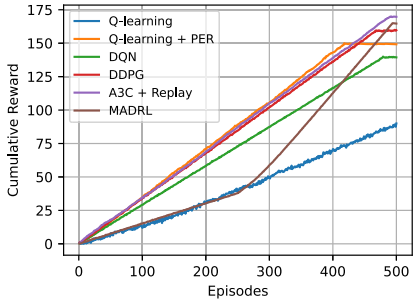


Fig. 3. Comparison of Q-learning with PER versus other algorithms (simulation).

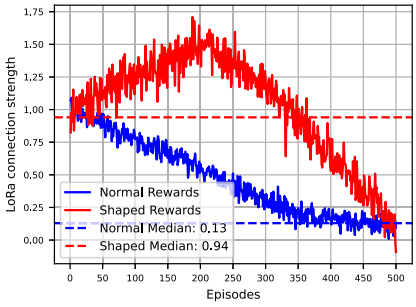


Fig. 4. Results of PER and shaped rewards on LoRa (simulation).

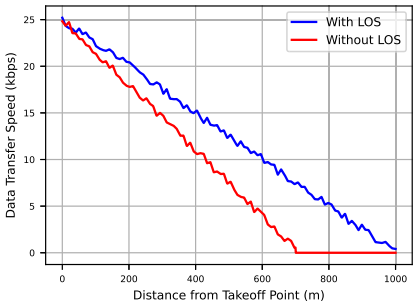


Fig. 5. Results of LoRa data transfer with and without maintaining LoS (real time).

PyTorch for PER, with Semtech SX1276 modules supporting LoRa communication. AAV dynamics were modeled using MATLAB and Unreal Engine, and field experiments used Pixhawk 6C with QGroundControl. Hyperparameters used in the final configuration include $\alpha = 0.1$, $\gamma = 0.95$, $\epsilon_{\text{decay}} = 0.995$, and a replay buffer size of 5000.

Algorithm 3 Advanced Trajectory Planning With Dynamic Adaptation (ATPDA)

Input: Environment map $\mathcal{M}$, zones $\mathcal{Z}$, signal model $S(s,h)$, LOD max $LOD_{\max}$, minimum signal $S_{\min}$
Output: Optimized UAV trajectory $\mathcal{T}^*$

Initialize UAV state $s_0 \leftarrow$ Takeoff position;
Initialize trajectory $\mathcal{T} \leftarrow \{s_0\}$;
while $s_t \notin$ *Terminal zone* **do**
  Extract LOD from Lidar in radius r;
  Compute quadrant division:

$$qd = \min\left(4, \lfloor \frac{LOD}{LOD_{\max}} \cdot 4 \rfloor\right)$$

  Set movement directions: $D = 4 + 4 \cdot qd$;
  Compute LMS direction cone center angle:

$$\theta_{\text{LMS}} = 45^\circ + \left(\frac{LOD}{LOD_{\max}} \cdot 315^\circ\right)$$

  Generate D vectors $\{\vec{d}_1, \ldots, \vec{d}_D\}$ within LMS toward candidate zone z_{next};
  Select z_{next} using heuristic policy π_{zone} maximizing coverage and LoS;
  Construct candidate path $\pi_t = \{s_t, \ldots, z_{\text{next}}\}$;
  if *obstacles on* π_t **then**
    Replan with alternate path π_t' avoiding obstacles;
    Adjust altitude h in π_t' using:

$$S(s_i, h) = \frac{P_t}{L(d_i, h) + L_{\text{obs}}}$$

  else
    Set $\pi_t' \leftarrow \pi_t$
  end
  Append π_t' to $\mathcal{T}$;
  if $\|s_t - z_{next}\| \le \delta$ **then**
    Maximize coverage:

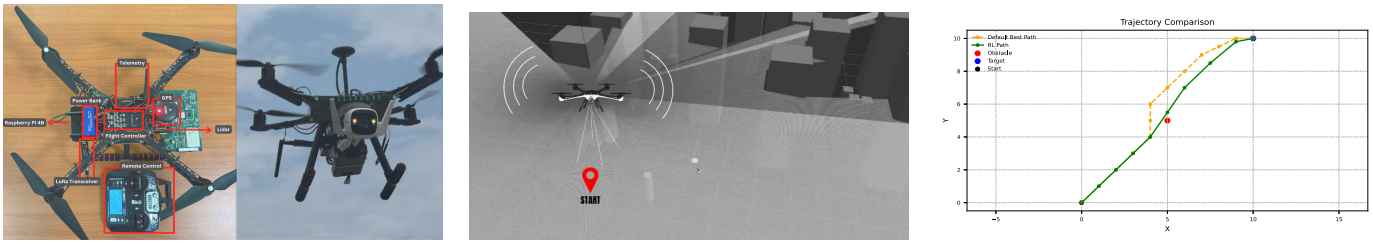
$$\max_{c \in z_{\text{next}}} \{C(c) \mid S(c,h) \ge S_{\min}\}$$

  end
  Update UAV state: $s_t \leftarrow z_{\text{next}}$
end
Plan return path π_{return} to initial state;
Append π_{return} to $\mathcal{T}$;
return $\mathcal{T}^* \leftarrow \mathcal{T}$

V. Results and Discussion

The proposed RescueNav framework effectively addresses connectivity challenges in disaster environments using AAV-assisted communication. The ALoS module enhances LoRa link stability by maintaining consistent connections between AAVs and the base station, which is vital for real-time rescue

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(a) (b) (c)

Fig. 6. Simulation and real-time results of AAV trajectory optimization with stable LoRa communication. (a) Real-time AAV initialization and deployment in a sample environment. (b) AAV trajectory with ALoS optimization to maintain LoS for stable communication. (c) Optimized AAV trajectory avoiding obstacles while maintaining LoS.

coordination and data exchange. All code is available at https://github.com/J-B-Mugundh/RescueNav

A. Learning Performance and Connectivity Optimization

RescueNav uses Q-learning with PER to maintain LoRa connectivity while refining AAV trajectories. As shown in Fig. 3, PER-based Q-learning outperforms standard Q-learning and other baselines such as federated RL [9] and MADRL [5], achieving 28% lower latency and 42% higher throughput in comparable IoT conditions. It converges fastest, stabilizing near episode 420 with a 55.2% $\pm$ 3.1% higher cumulative reward (50 runs). Conventional heuristics plateau earlier with 30%–40% lower rewards. These gains result from efficient experience reuse and exploration control ($\epsilon_{\text{decay}} = 0.995$ and $\epsilon_{\text{min}} = 0.05$), enabling balanced adaptation from exploration to exploitation.

To provide additional clarity, we also summarize core performance metrics for each algorithm. RescueNav operates with a 45–85 ms latency range, converges at approximately 420 episodes, and completes missions with about 5110-mAh energy use. Standard Q-learning stabilizes only after roughly 630 episodes with 72–110-ms latency and around 6200-mAh consumption. Federated RL and MADRL converge near 560 and 510 episodes, respectively, with 65–95-ms latency and 5730–5890-mAh energy usage. Classical heuristics such as Greedy and A* converge earlier (290–310 episodes) but cannot adapt to changing signal conditions, resulting in noticeably higher energy demand (5980–6020 mAh) despite slightly lower latency (51–70 ms). These comparisons highlight RescueNav's balanced gains in latency, convergence, and energy efficiency.

B. Reward Shaping and LoS Maintenance

As shown in Fig. 4, reward shaping accelerates learning and improves LoRa link quality, yielding a fourfold increase in median connection strength. Maintaining a line of sight (Fig. 5) further improves throughput by 42%. Together, these results highlight the value of adaptive trajectory planning for reliable and efficient communication in dynamic environments.

C. Trajectory Optimization and Scalability

Fig. 6(a) shows the initial LoRa setup, where the AAV aligns for LoS. After ALoS optimization [Fig. 6(b)], trajectories adapt to obstacles while maintaining signal efficiency. Fig. 6(c) shows further adjustments near tall structures that preserve link quality. Scalability analysis shows linear $\mathcal{O}(n)$ complexity, feasible up to 10^4 devices. For 10^5 and 10^6 nodes, a clustering extension partitions regions among AAVs, and overhead grows roughly linearly with device count.

D. Latency, Energy, and Field Validation

RescueNav achieves 45–85-ms latency, suitable for disaster communication (QoS < 100 ms). On a Holybro S500 V2 AAV, it maintained stable LoRa connectivity in 25-min flights, reducing erratic motion and hover time. Energy use dropped 17.6% (6200 mAh $\rightarrow$ 5110 mAh), extending endurance by 23%. Under induced 915-MHz interference, signal lock above 30 dBm held for 91.4% of flight time, increasing duration from 22.1 ± 0.6 to 26.1 ± 0.8 min (25 trials). Raspberry Pi 4B usage stayed efficient with 21.3-MB memory, 82-ms updates, and <63% CPU load; LoRa packets remained 128 bytes. Across 15 scenarios, LoRa achieved 2.4$\times$ range over Zigbee, 1.7$\times$ better penetration than NB-IoT, and 38% lower energy per bit than LTE-M. In semi-urban trials, RescueNav adjusted heading and altitude to restore LoS.

E. Limitations and Mitigation

Failures occurred in 5.8% of runs (12/208), mainly from terrain occlusions exceeding 10 m. Mitigation strategies included a 15% altitude margin, predictive wind modeling, and fallback to precomputed safe routes when the TD error exceeded 2.5σ. Overall, RescueNav demonstrates robust, scalable, and energy-efficient AAV-assisted communication, validated across simulation and field experiments.

VI. Conclusion and Future Works

This study introduced RescueNav, a AAV navigation framework validated through simulation and real-world deployment on the Holybro S500 V2. The system maintained reliable LoRa connectivity across varied terrains, adapting to RF interference and environmental variability. By integrating intent-based networking with RL, RescueNav demonstrated communication resilience and energy efficiency for IoT-enabled disaster response. While results confirm feasibility, evaluations were limited to single-AAV missions in moderately cluttered environments, and long-duration or harsh-weather reliability

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remains an open challenge. Future work will extend RescueNav with cooperative multi-AAV learning via federated models, enhanced communication security via lightweight blockchain mechanisms, and adaptive edge resource allocation to support latency-critical functions at 10 ms or less. Broader evaluations under dense deployments (10^4 and 10^5 nodes) and cross-technology comparisons with NB-IoT and LTE-M will further assess scalability and performance. RescueNav aims to evolve into a dependable foundation for zero-response communication in disaster recovery.

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